Lecture-20
Dimensional Analysis I:
Dimensionless forms of conservation equations

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Learning Objectives

By the end of this lecture, you should be able to

1. Distinguish between **counted variables and measured variables**.
2. Understand that no **natural or absolute units** exist for physical variables.
3. Understand the concept of **dimensional homogeneity**.
4. Transform the **Navier-Stokes equations for incompressible flow** into their **dimensionless form.**

Fundamental ideas in dimensional analysis

- Any quantity that is measured typically has a **dimension** associated with it.
- **No natural or fundamental units** of measurement exist for a physical variable.
- Physical equations can be regarded as **relations between values of variables**, expressed in consistent units.
- Any physical equation is **dimensionally homogeneous**.

Dimensionless form of the iNS equation

- We can use **relevant scales in a problem to transform** the Navier-Stokes equations for incompressible flow.

$$x^* = \frac{x}{L}, \quad y^* = \frac{y}{L}, \quad u^* = \frac{u}{V_\infty}, \quad v^* = \frac{v}{V_\infty}, \quad \text{and} \quad p^* = \frac{p}{\rho V_\infty^2}$$

$$\frac{\partial u^*}{\partial x^*} + \frac{\partial v^*}{\partial y^*} = 0$$

$$u^* \frac{\partial u^*}{\partial x^*} + v^* \frac{\partial u^*}{\partial y^*} = -\frac{\partial p^*}{\partial x^*} + \frac{\mu}{\rho V_\infty L}\left(\frac{\partial^2 u^*}{\partial x^{*2}} + \frac{\partial^2 u^*}{\partial y^{*2}}\right)$$

$$u^* \frac{\partial v^*}{\partial x^*} + v^* \frac{\partial v^*}{\partial y^*} = -\frac{gL}{V_\infty^2} - \frac{\partial p^*}{\partial y^*} + \frac{\mu}{\rho V_\infty L}\left(\frac{\partial^2 v^*}{\partial x^{*2}} + \frac{\partial^2 v^*}{\partial y^{*2}}\right)$$

Fox, Robert W., Alan T. McDonald, and John W. Mitchell. 2020. *Fox and McDonald's Introduction to Fluid Mechanics*. 10th ed. Hoboken, NJ: John Wiley & Sons.

Revisiting the learning objectives

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